

Temporal Connectivity in Landscape Genetics: From Static Graphs to Time-Series Genomics

A Literature Review Generated with AI Assistance

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Abstract

Landscape genetics has traditionally treated connectivity as a static property inferred from a single snapshot of genetic variation. Temporal genomic datasets — historical specimens, resampled populations, and time-series of allele frequencies — now allow connectivity itself to be estimated as a dynamic process. This review synthesizes methods linking landscape resistance surfaces to temporal genetic differentiation, including time-series F_{ST} approaches, effective migration surface models fit to based inference for resistance parameterization. We discuss how temporal designs disentangle demographic change from loss responses, and forecast gene flow under future land-use scenarios.

Keywords: landscape genetics, gene flow, temporal genomics, connectivity.

Placeholder Notice

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Scope

The forthcoming review will synthesize recent literature on the topic above, covering its principal methods, findings, open challenges, and directions for future work within the field of Ecology & Evolution.

Data and Code Availability

No new data or code are associated with this placeholder.

References

- [1] LitReview Consortium (2026). *LitReview: an arXiv-style repository of AI-generated literature reviews for the basic sciences*. ai.litreview.org.